FINANCIAL KPI MONITORING & PROFITABILITY DASHBOARD

Power BI Project Summary

Tool	Microsoft Power BI Desktop
Data source	Excel / CSV sample dataset
Dataset size	180 fictional transaction rows · 20 columns
Currency	Indian Rupees (INR)

Project Overview

This project demonstrates how Power BI can transform transaction-level financial data into an interactive dashboard for monitoring revenue, costs, profitability, and budget performance. It is designed as a beginner-friendly portfolio project that demonstrates data preparation, data modeling, DAX measures, visual design, and business interpretation.

Objective: help business users answer where revenue is generated, which product categories and channels are profitable, whether revenue targets are being achieved, and how profitability changes over time.

Business Questions

• Which regions generate the highest revenue?

• Which product categories contribute the most net profit and margin?

• Is actual revenue meeting the budget target?

• How does net profit margin change month by month?

• Which sales channels perform better on revenue and profitability?

Dataset and Preparation

The sample workbook contains 180 fictional transaction-level rows and 20 columns, including date, region, department, product category, revenue, cost of goods sold, operating expenses, gross profit, net profit, profit margin, budgeted revenue, actual revenue, budget variance, customer segment, and sales channel.

Preparation workflow: import the Excel workbook into Power BI, inspect and correct data types in Power Query, check for missing values and duplicates, apply changes, and use a clearly named fact table (FactFinancials). A date dimension (DimDate) supports chronological analysis and time-based measures.

Data Model and Measures

The intended model uses FactFinancials for transaction-level values and DimDate for calendar analysis. Additional dimensions such as product, region, department, customer segment, and sales channel can be created to support a star schema.

Measure	Purpose
Total Revenue	Sum of recorded revenue
Total Cost	COGS plus operating expenses
Gross Profit	Revenue minus COGS
Net Profit	Gross profit minus operating expenses
Net Profit Margin %	Net profit divided by revenue
Budget Variance	Actual revenue minus budgeted revenue
Budget Achievement %	Actual revenue divided by budgeted revenue
Revenue MoM Growth %	Change in revenue compared with the previous month

Dashboard Design

The report is designed around a clean corporate layout using blue for primary financial metrics, green for favorable outcomes, red for negative outcomes, orange for attention items, and a light neutral background. Use consistent typography, aligned visuals, clear labels, INR currency formatting, and percentage formatting for margin measures.

Visual	Fields / purpose
KPI cards	Total Revenue, Total Cost, Gross Profit, Net Profit, Net Profit Margin %, Budget Achievement %
Column chart	Revenue by Region
Bar chart	Net Profit by Product Category
Line chart	Monthly Total Revenue and Net Profit trend
Donut chart	Revenue distribution by Sales Channel
Matrix	Region with Revenue, Cost, Net Profit, and Net Profit Margin %
Slicers	Region, Department, Product Category, Sales Channel, Month
Date slicer	DimDate[Date] with Between selection

Business Insights from the Sample Data

The following figures were reported from analysis of the supplied fictional sample dataset and should be checked against the final Power BI report and its active filters before publication:

• **Budget:** Budget achievement is approximately 98.99%, indicating actual revenue is about 1.01% below the practice target.

• **Regions:** North was reported as the highest-revenue region; West was reported as the lowest-revenue region. South was reported to have a higher net profit margin than North.

• **Products:** Electronics was reported as the highest net-profit category; Accessories was reported as having the highest net profit margin.

• **Monthly margin:** May was reported as the lowest-margin month (about 16.45%) and August as the highest (about 23.85%).

• **Channels:** Direct Sales was reported as the largest channel by net profit; Retail Partner was reported as having the highest net profit margin.

Strategic Use

• Allocate resources by considering both revenue and profitability, not revenue alone.

• Review pricing, product mix, COGS, and operating expenses where margins are weak.

• Investigate revenue shortfalls against budget and improve forecasting assumptions.

• Compare channels by net profit and margin before changing marketing or sales investment.

• Use slicers and date filters to investigate changes by period, region, department, category, and channel.

Limitations and Notes

The dataset is synthetic and intended for learning and demonstration. Actual Revenue equals Revenue in the supplied sample. Budget targets are synthetic practice values and are not a production finance budget model. Marketing spend and customer acquisition costs are not separately captured, so marketing ROI cannot be directly calculated. Confirm all insight values in the completed PBIX before using them in a public portfolio.